

1.

 **SIMATS**
ENGINEERING

SIMATS Online Programming Lab
C++ PROGRAMMING

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QUESTIONS 9 / 20 completed

Geometry Figures ▾

GEOMETRY FIGURES

Geometry Figures:
Build a class hierarchy for various geometric figures (circle, rectangle, triangle) with respective properties and methods to calculate area and perimeter.
Sample Input:
circle_r=5
rectangle l = 4, w = 6

PROGRAM EDITOR C++ ▾

Run

Save

```
1 #include <iostream>
2 #include <cmath>
3 using namespace std;
4
5 class Shape {
6 public:
7     virtual double area() = 0;
8     virtual double perimeter() = 0;
9 };
10
11 class Circle : public Shape {
12     double r;
13 public:
14     Circle(double radius) : r(radius) {}
15     double area() override {
16         return M_PI * r * r;
17     }
18     double perimeter() override {
19         return 2 * M_PI * r;
20     }
21 }
```

INPUT

5
4 6
3 4 3 5simatslab.saveetha.com/practice_app/dashboard_student

2.



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Library Manage

Library Management System

Library Management System: Create a class for a library system that manages books, including attributes like book name, author, and availability. Implement functions for issuing and returning books. Sample Input:
// Creating an instance of the

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 #include <vector>
3 #include <string>
4
5 using namespace std;
6
7 class Book {
8 public:
9     string title;
10    string author;
11    bool isAvailable;
12
13    Book(string t, string a) : title(t), author(a), isAv
14 };
15
16 class Library {
17 private:
18     vector<Book> books;
19
20 public:
```

INPUT

Books in the library:
Book Name: The Great Gatsby, Author: F. Scott Fitzgerald, Availability: Available
Book Name: To Kill a Mockingbird, Author: Harper Lee, Availability: Available

Books in the library:
Book Name: To Kill a Mockingbird, Author: Harper Lee, Availability: Available

Books in the library:
Book Name: The Great Gatsby, Author: F. Scott Fitzgerald, Availability: Available
Book Name: To Kill a

3.



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QUESTIONS 9 / 20 completed

Car Rental System

CAR RENTAL SYSTEM

Car Rental System:
Design a class to represent cars for a rental service, including details like model, availability, and rental cost per day. Implement methods for booking and returning cars.
Sample Input:
Enter car model:
Toyota Camry

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 #include <string>
3
4 using namespace std;
5
6 class Car {
7 private:
8     string model;
9     double rentalCostPerDay;
10    bool isAvailable;
11
12 public:
13     Car(string m, double cost) {
14         model = m;
15         rentalCostPerDay = cost;
16         isAvailable = true;
17     }
18
19     void displayDetails() {
20         cout << "Available Cars in the Rental System:" <
```

INPUT

Toyota Camry
50

33°C

Search

10

4.



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QUESTIONS 9 / 20 completed

Music Player

MUSIC PLAYER

Music Player:
Construct a class representing a music player with functionalities for playing, pausing, and stopping songs.
Include attributes like song name and artist.
Sample Input:
Enter song name:
Shape of You
Enter artist name: Ed

PROGRAM EDITOR
C++
Run Save

```

1 #include <iostream>
2 #include <string>
3
4 using namespace std;
5
6 class MusicPlayer {
7 private:
8     string songName;
9     string artistName;
10
11 public:
12     void playSong(string song, string artist) {
13         songName = song;
14         artistName = artist;
15         cout << "Now playing: " << songName << " by " <<
16     }
17
18     void pauseSong() {
19         cout << "Song paused" << endl;
20     }
21

```

INPUT

Shape of You
Ed Sheeran
pause
stop

5.



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QUESTIONS 9 / 20 completed

Inventory Manager

INVENTORY MANAGEMENT

Inventory Management:
Develop a class for inventory management in a store, keeping track of items, their quantity, and methods for adding or removing stock.
Sample Input:
Adding new item to inventory.
Enter item name:
Laptop
Enter quantity: 10

PROGRAM EDITOR
C++
Run Save

```

1 #include <iostream>
2 #include <string>
3
4 using namespace std;
5
6 class Inventory {
7 private:
8     string itemName;
9     int quantity;
10
11 public:
12     Inventory() : itemName(""), quantity(0) {}
13
14     void addItem(string name, int qty) {
15         itemName = name;
16         quantity += qty;
17         cout << "Inventory updated." << endl;
18         cout << "Item: " << itemName << ", Quantity: " <
19     }
20
21

```

INPUT

Laptop
10
Laptop
3